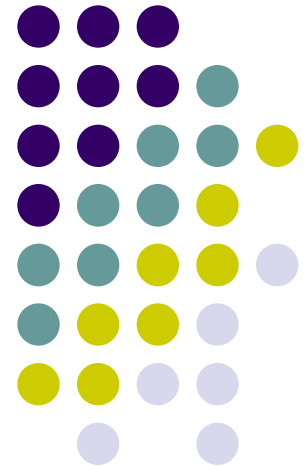
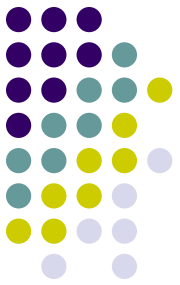


Developing simple Excel models of physiological processes

Dynamic Modelling for
Human-centered Systems

Lecture 2

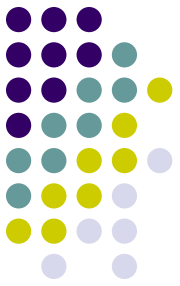




Course outline

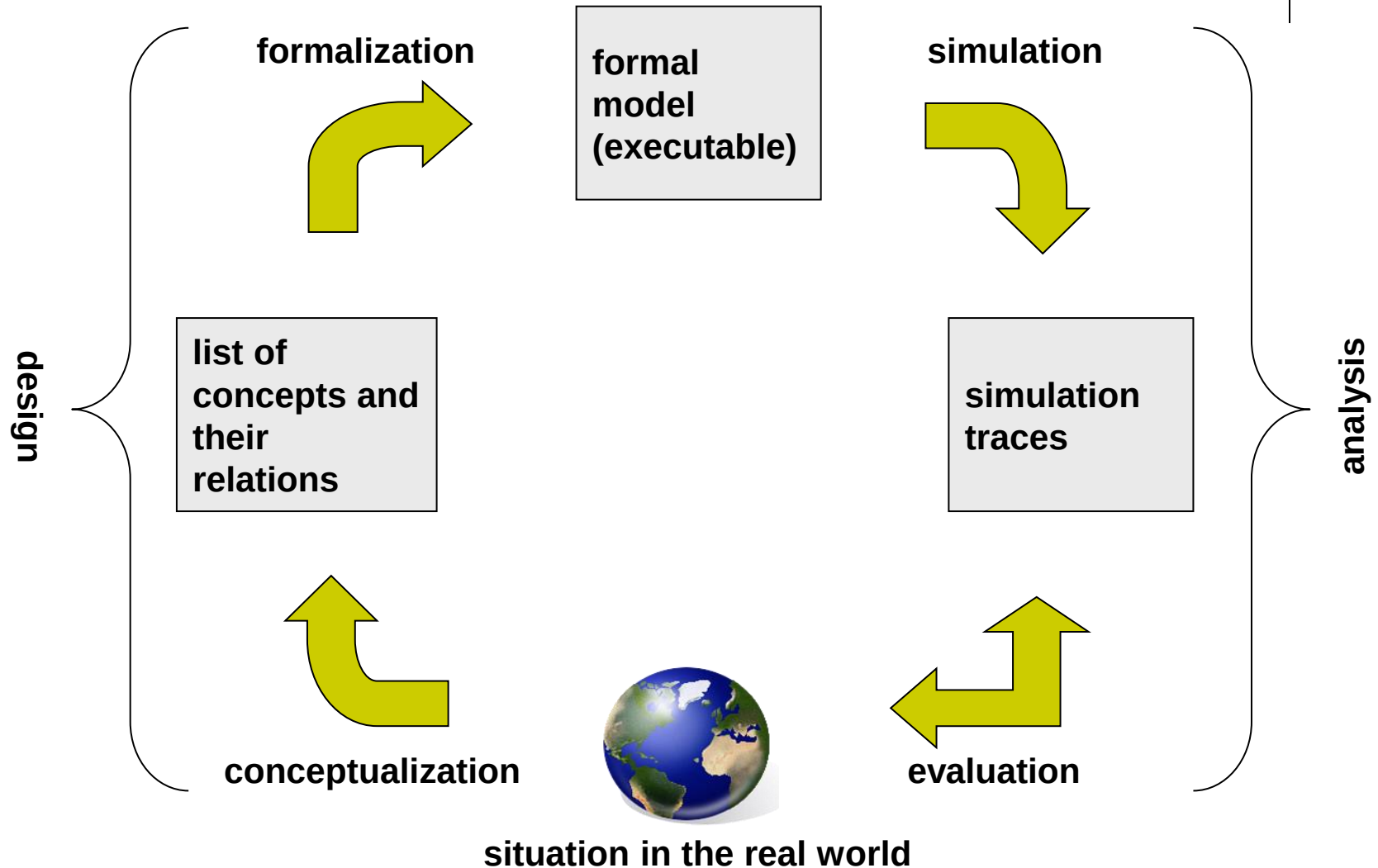
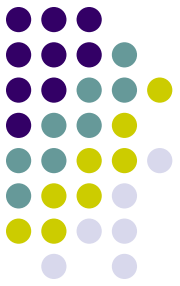
- This week: what is dynamic modelling?
 - today: **creating models of physiological processes**
- Next week: modelling social processes
- Week 3: modelling behaviour
- Week 4: creating domain models
- Week 5: agent-based modelling
- Week 6: analysis and support models
- Week 7: wrap-up

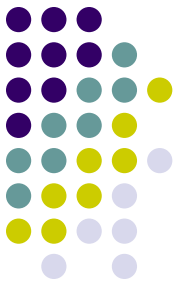
Which processes will we model?



- Food intake of bacterium
- Responses of sea hare on stimuli
- At the same time:
 - rudimentary decision making process
 - neurally-inspired learning process
- What we will learn:
 - how to create models, i.e. conceptualize theoretical knowledge
 - building blocks for other models

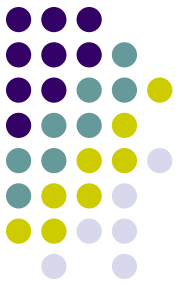
Modelling and Simulation Cycle





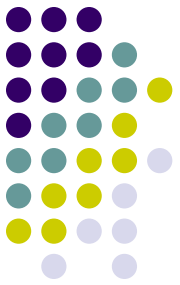
Steps

1. **Conceptualisation:** identification of important concepts and relations
2. **Formalisation:** describe concepts and relations in an unambiguous way
3. **Simulation:** running experiments with models to generate simulations traces
4. **Evaluation:** verify whether the simulations are in line with expectations (properties)



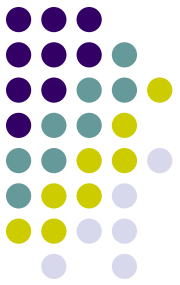
Creating models in practice

- Knowledge is basis
- Similar approach for all models in syllabus:
 - study domain literature (syllabus section 1)
 - identify main aspects + relations (syllabus section 2)
 - also “characteristic patterns”
 - create formal representation (syllabus section 3)
 - perform simulations (syllabus section 4)
 - evaluate simulations (syllabus section 5)



Syllabus: Chapter 3

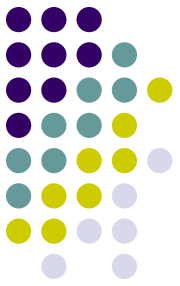
MODELLING FOOD INTAKE BY E-COLI



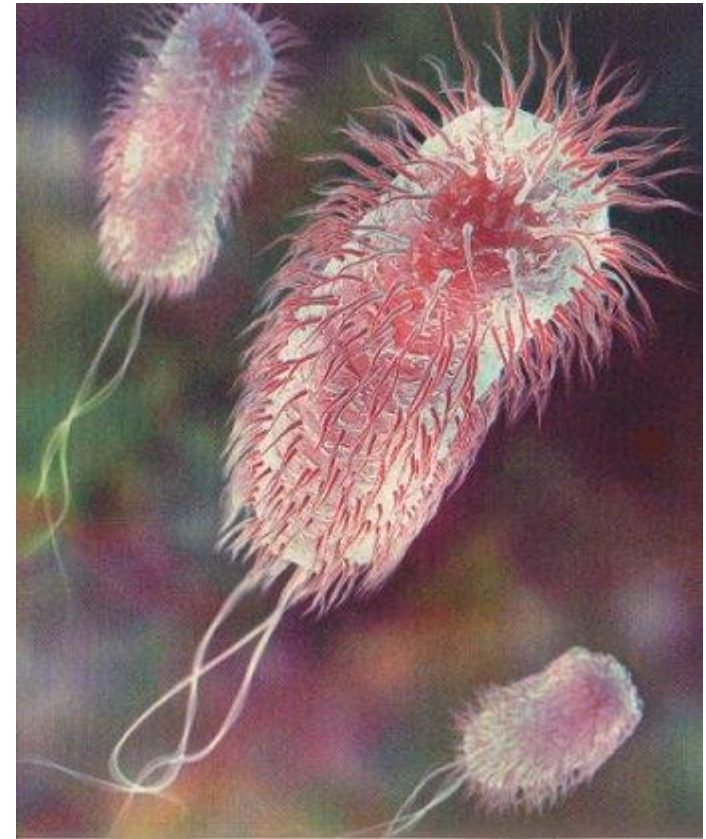
Modelling decision making

- How does *decision making* work?
- Complex process
 - much research about patterns
 - psychological experiments
 - decision models
 - underlying mechanism are difficult to study
 - *brain imaging* only shows activity of brain areas
 - bacterium is simple enough to understand chemical pathways

Bacterial food intake



- E-Coli: bacterium
 - located in intestines
 - consumes glucose or lactose

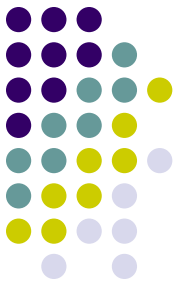


Behaviour bacter

Assignment

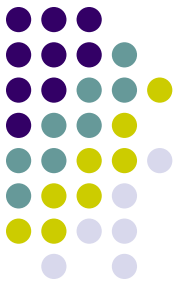
- Which pattern do you see in the behaviour of the bacterium?
- Try to formulate it as a rule that holds for all time points

	0	1	2	3	4	5	6
observa tion	glucose	glucose lactose	lactose	lactose	glucose lactose	glucose lactose	glucose lactose
action		import glucose	import glucose	import lactose	import lactose	import glucose	import glucose



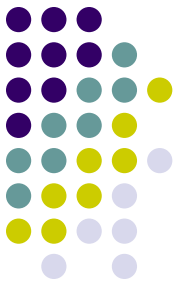
Pattern

- Food intake behaviour:
 - At any point in time
if it observes that - at previous time point -
glucose is present,
then it will import glucose
 - At any point in time,
if it observes that - at previous time point –
lactose is present,
and that no glucose is present,
then it will import lactose



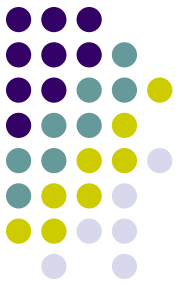
Internal mechanism

- Bacterium can “observe” the environment (sensing) and “decide” about
 - **lactose** intake, or
 - **glucose** intake
- Intake of lactose or glucose is controlled by two specific ***enzymes***
- “Lactose intake enzyme” is produced if:
 - chemical substance CRPcAMP is present, and
 - (low) concentration of lactose in cell



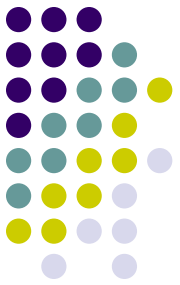
Internal mechanism (2)

- Substances in cell:
 - CRPcAMP is *only* present within cell if glucose is <0.1 mmol/l outside the cell
 - indicator of *absence* of glucose
 - a little bit of lactose is present in cell when there is lactose outside the cell (infiltrates)
- The bacterium will grow from both lactose intake as glucose intake



Characteristic patterns

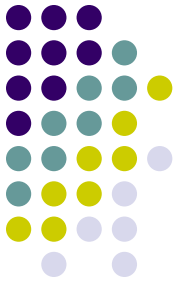
- If food is present, then the bacterium will grow
- If lactose is externally present, but not glucose, then lactose is imported,
- If glucose is externally present but not lactose, then glucose is imported,
- If both glucose and lactose are externally present, then glucose is imported but not lactose.



Assignment

- Which concepts do you think are relevant?
 - Make a list of concepts
 - *3 minutes*
 - *groups of three; I will ask a few of you to come up with suggestions*
-
- Important question:
 - what do we consider as concepts?
 - it can help to make a categorization

Concepts



External state

-
-

Internal states reflecting the external state

-
-

Internal states for action initiation

-
-

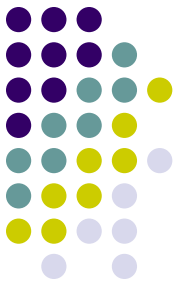
Action performance

-
-

Internal state showing the effect of actions

-
-
-

Concepts



External state

- glucose externally present
- lactose externally present

Internal states reflecting the external state

- CRPcAMP internally absent (glucose indicator)
- some lactose internally present (lactose indicator)

Internal states for action initiation

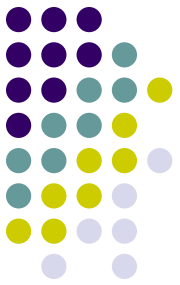
- enzyme for glucose import present
- enzyme for lactose import present

Action performance

- glucose import takes place
- lactose import takes place

Internal state showing the effect of actions

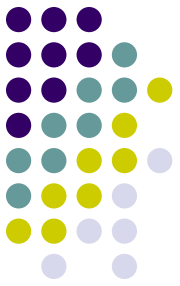
- substantial glucose internally present
- substantial lactose internally present
- mass



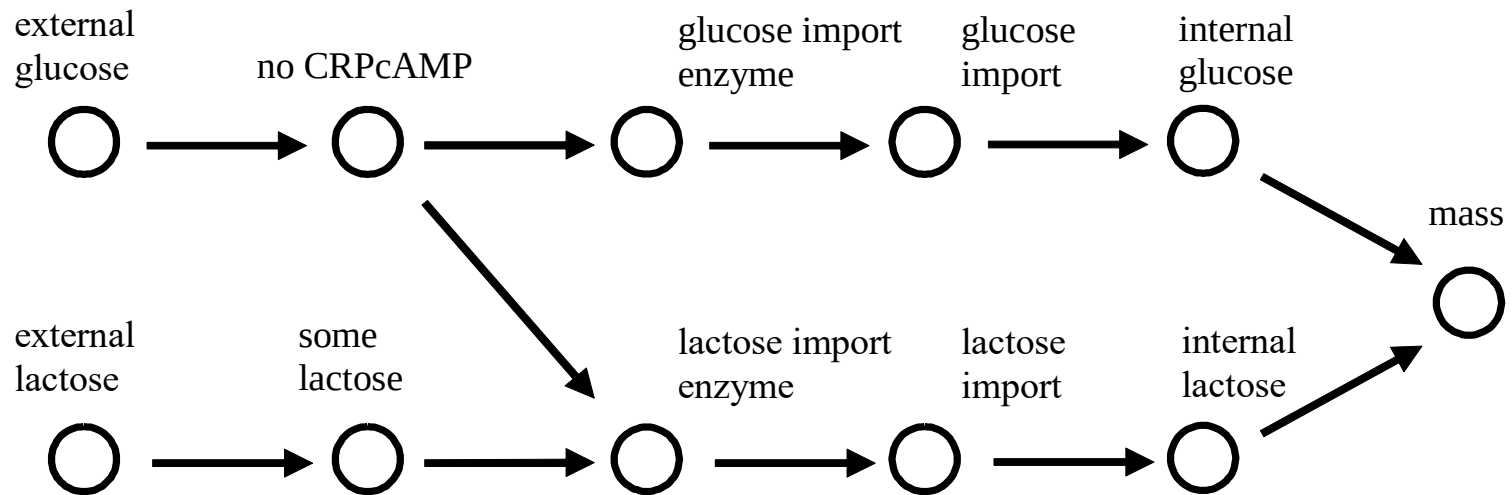
Assignment

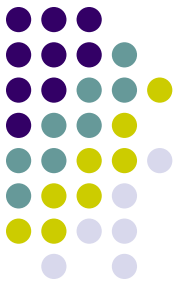
- Which direct relations can be identified between the concepts?
- Draw a diagram with circles for concepts and arrows between concepts if a concept affects another concept.
- *we will construct it together based on your input*

Conceptualisation (2)



- “Affects” relations between concepts

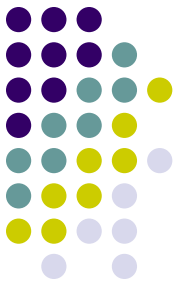




Formalisation of concepts

- Variables for every concept

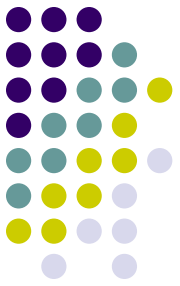
- glucose externally present: *EXTgluc* [0/1]
- lactose externally present: *EXTlac* [0/1]
- glucose indicator (no CRPcAMP): *glucIND* [0/1]
- lactose indicator: *lacIND* [0/1]
- glucose import enzyme: *glucimpENZ* [0/1]
- lactose import enzyme: *lacimpENZ* [0/1]
- glucose internally present: *INTgluc* [0/1]
- lactose internally present: *INTlac* [0/1]
- mass: *M* [real]



Formalisation of relations

- Example relations:
 - From external glucose to glucose indicator
 - $glucIND(t+\Delta t) = EXTgluc(t)$
 - From lactose and glucose indicator to enzyme for lactose import
 - $lacimpENZ(t+\Delta t) = lacIND(t) * (1 - glucIND(t))$
 - Effect on mass
 - $M(t+\Delta t) = M(t) + \gamma (INTgluc(t) + INTlac(t)) * M(t)$
- *Almost logical representation*

Formalisation (2)

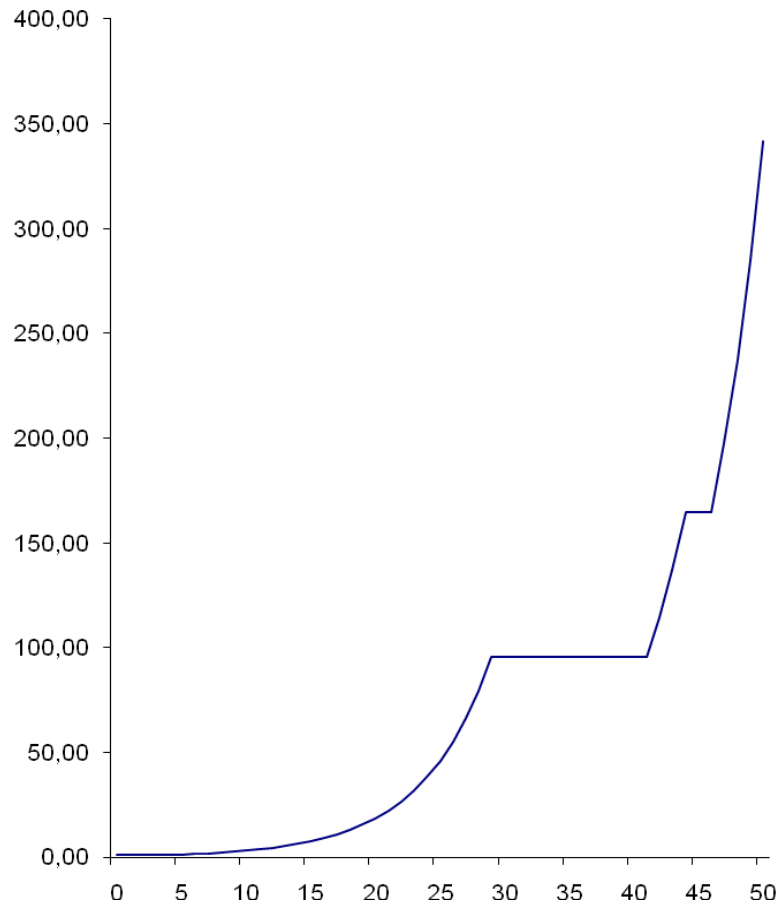
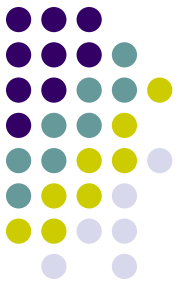


- Dependencies

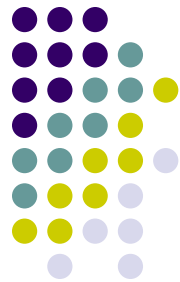
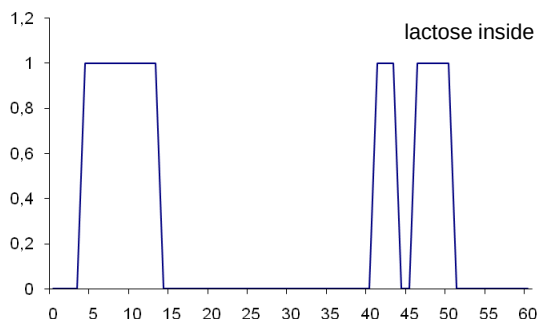
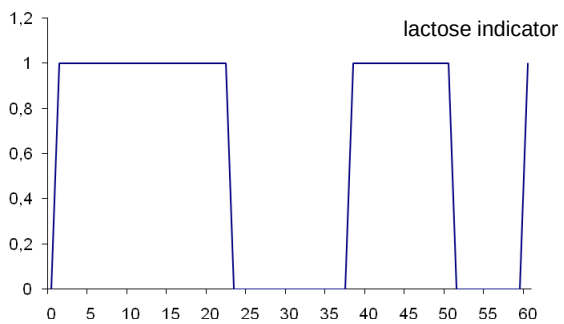
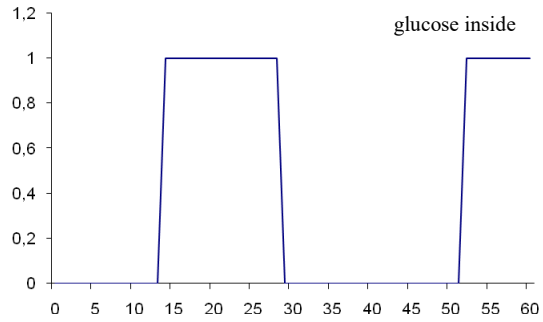
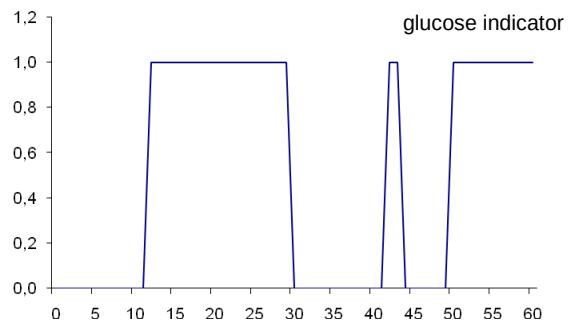
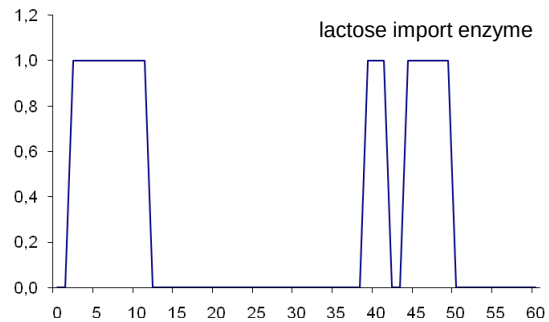
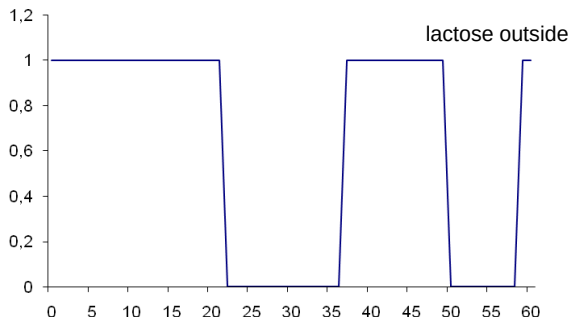
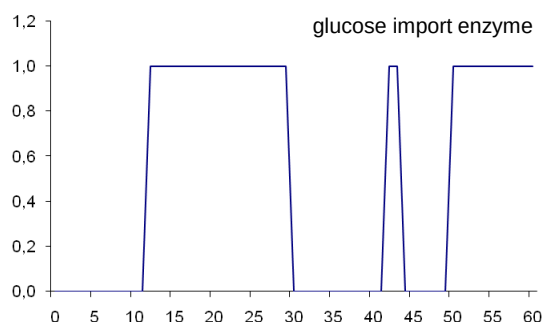
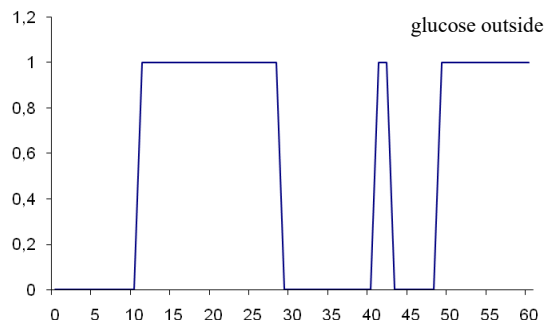
t	int gluc	int lact	mass	add mass	γ
0	1	0	1	0.20	0.2
1	0	1	1.2	0.24	0.2
2	0	1	1.44	0.29	0.2
3	0	1	1.73	0.35	0.2
4	0	1	2.08	0.42	0.2
5	0	1	2.50	0.50	0.2

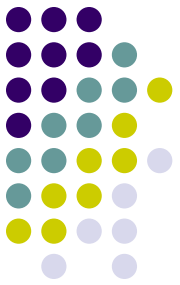
The diagram shows dependency arrows between the variables in the table across time steps. Arrows indicate that the 'mass' and 'add mass' values at time t depend on the 'int gluc' and 'int lact' values at time t and the 'mass' value at time $t-1$. Specifically, at $t=0$, 'int gluc' and 'int lact' depend on their own values at $t=0$. At $t=1$, 'mass' and 'add mass' depend on 'int gluc' and 'int lact' at $t=1$ and 'mass' at $t=0$. This pattern continues for subsequent time steps.

Simulation bacteria growth



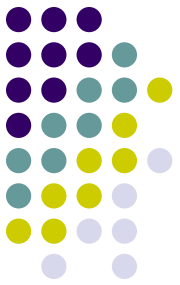
Simulation experiments





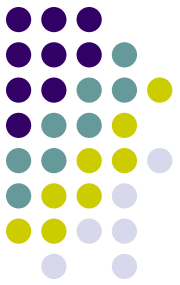
Evaluation

- Characteristic patterns:
 - If food is present, then the bacterium will grow
 - If lactose is externally present, but not glucose, then lactose is imported,
 - If glucose is externally present but not lactose, then glucose is imported,
 - If both glucose and lactose are externally present, then glucose is imported but not lactose.



Summary

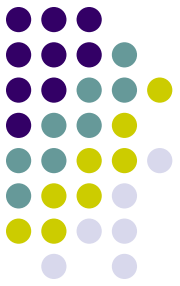
- Food intake by bacteria can be modelled numerically
 - extensions with role of DNA and mRNA are possible
= see reader
- Decision making can be modelled by:
 - creating internal representations of external situation
 - combine different internal concepts in a “smart” way



Syllabus: Chapter 4

NEURAL MECHANISM FOR LEARNING

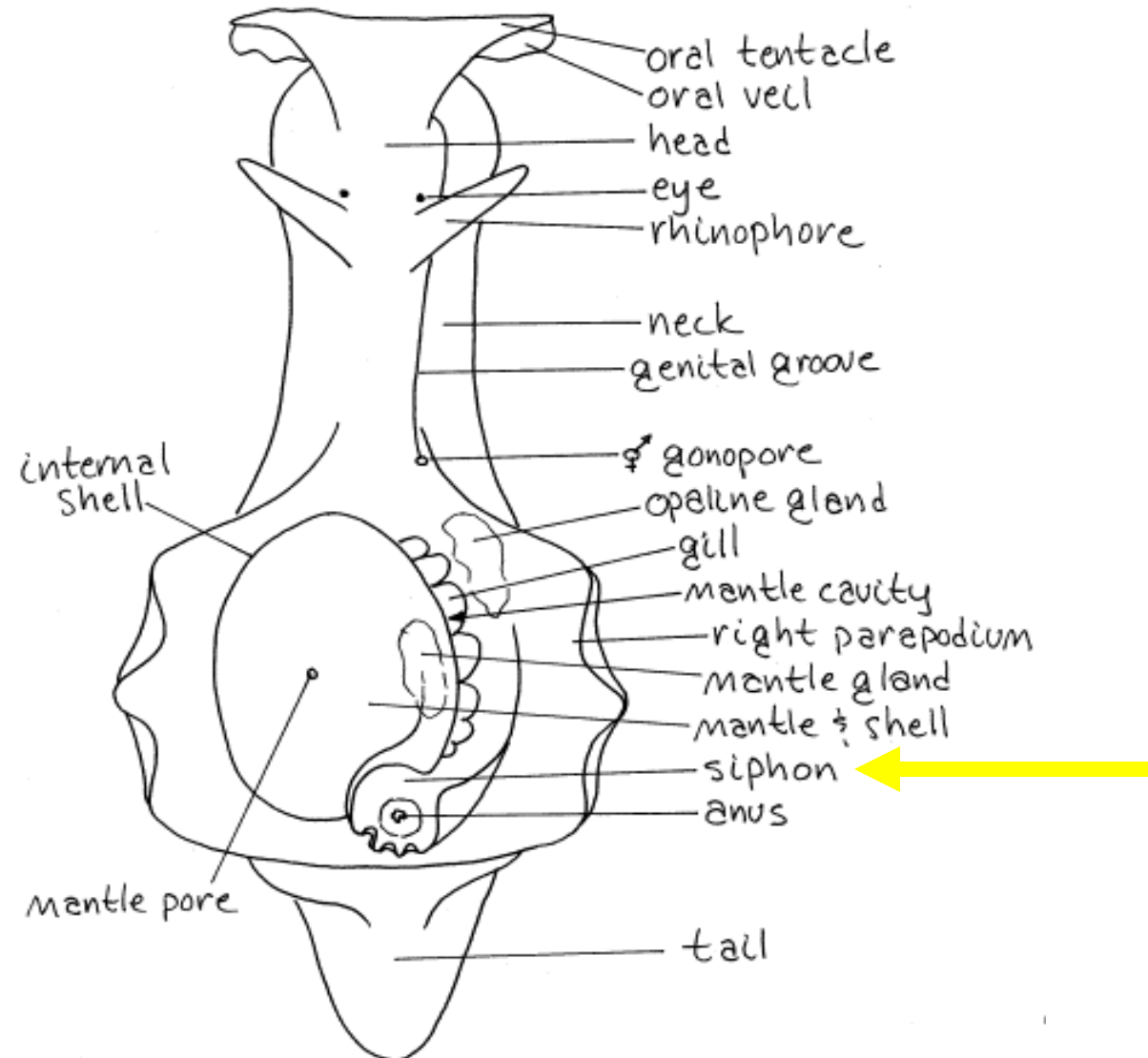
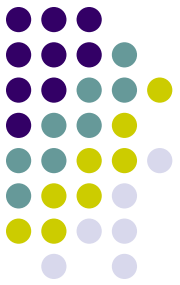
Aplysia



- Aplysia is a sea hare that is often used to do experiments.
 - for example, it performs classical conditioning: learning to move upon a siphon touch.

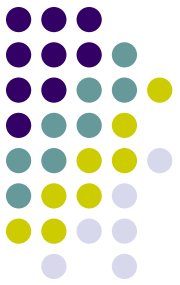


Aplysia



- The siphon is a spot on the back of the animal.

Behaviour before Learning Phase



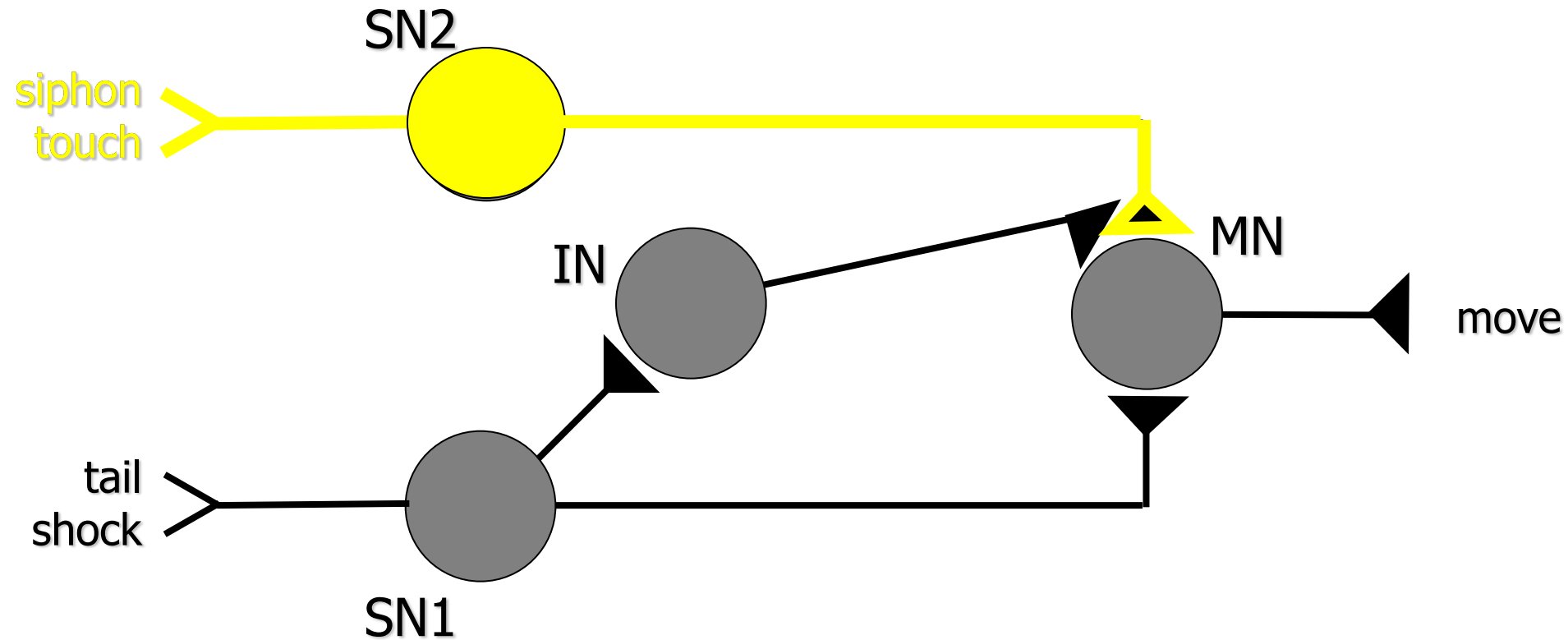
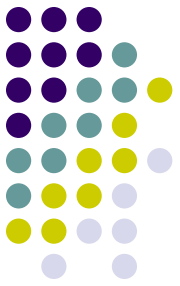
- Stimuli:
 - provide light tail shock
 - provide light touch on siphon
- Initially the following behaviour is shown:
 - a tail shock leads to a response (moving: contraction)
 - a light touch on its siphon does not trigger such a response

Behaviour after a Learning Phase

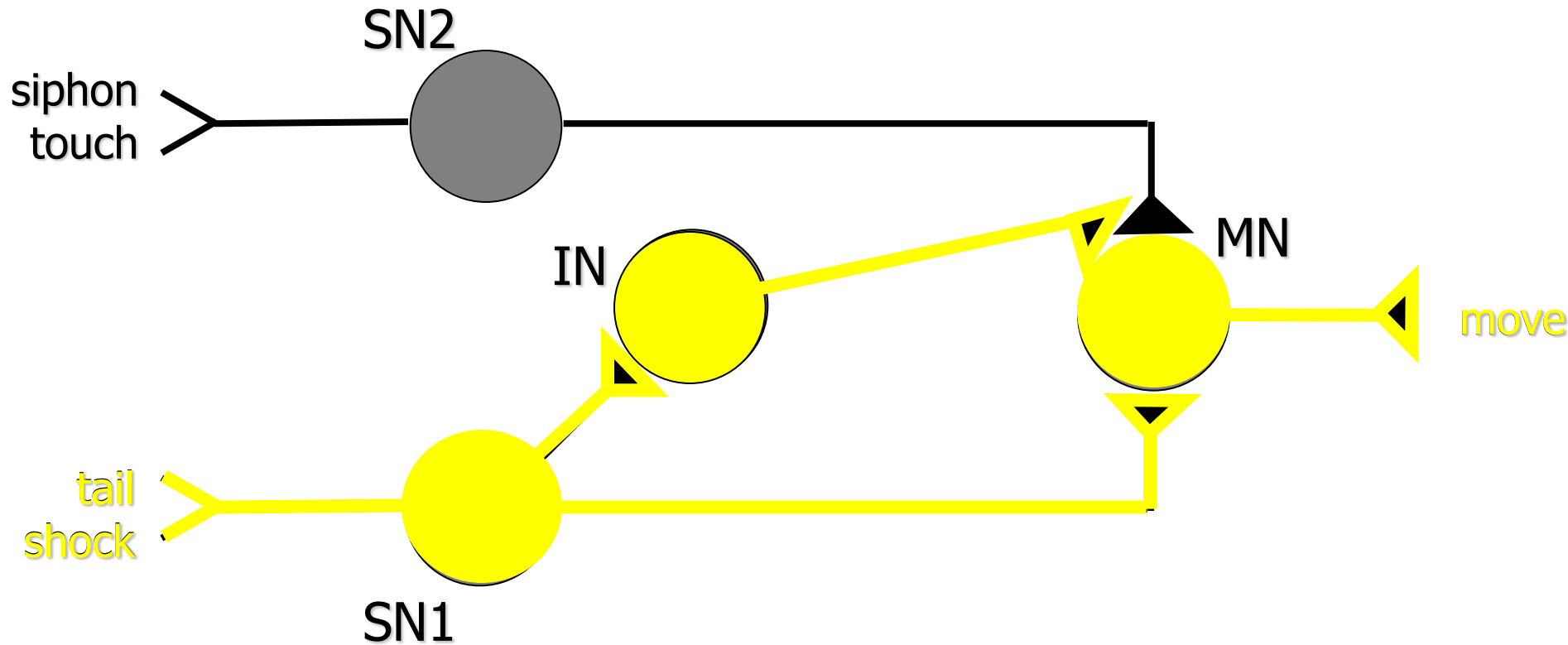
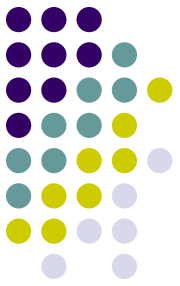


- It turns out that after several experiments the behaviour has changed (**adapted** behaviour):
 - the animal also responds (moving) on the siphon touch

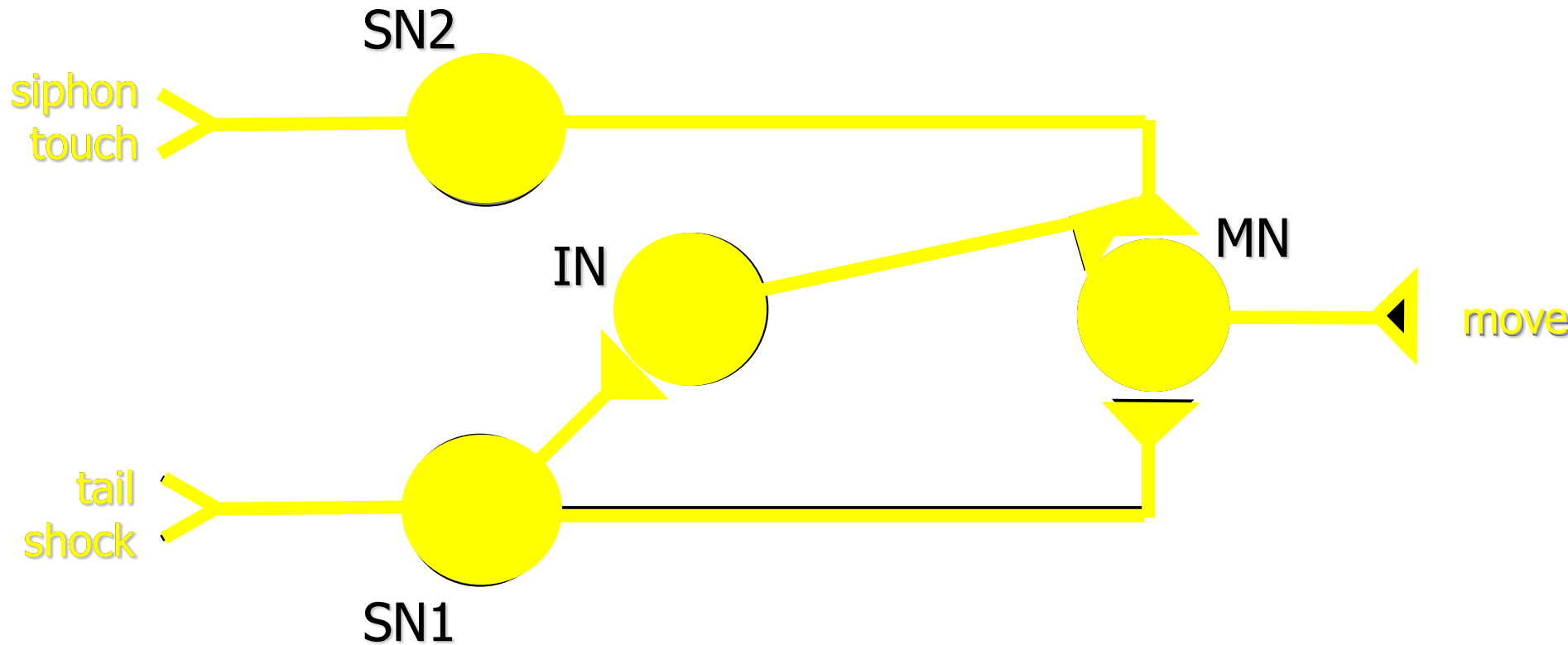
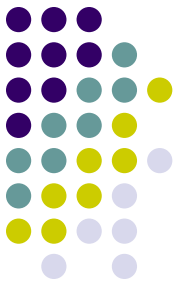
Neural Mechanism: Siphon touch



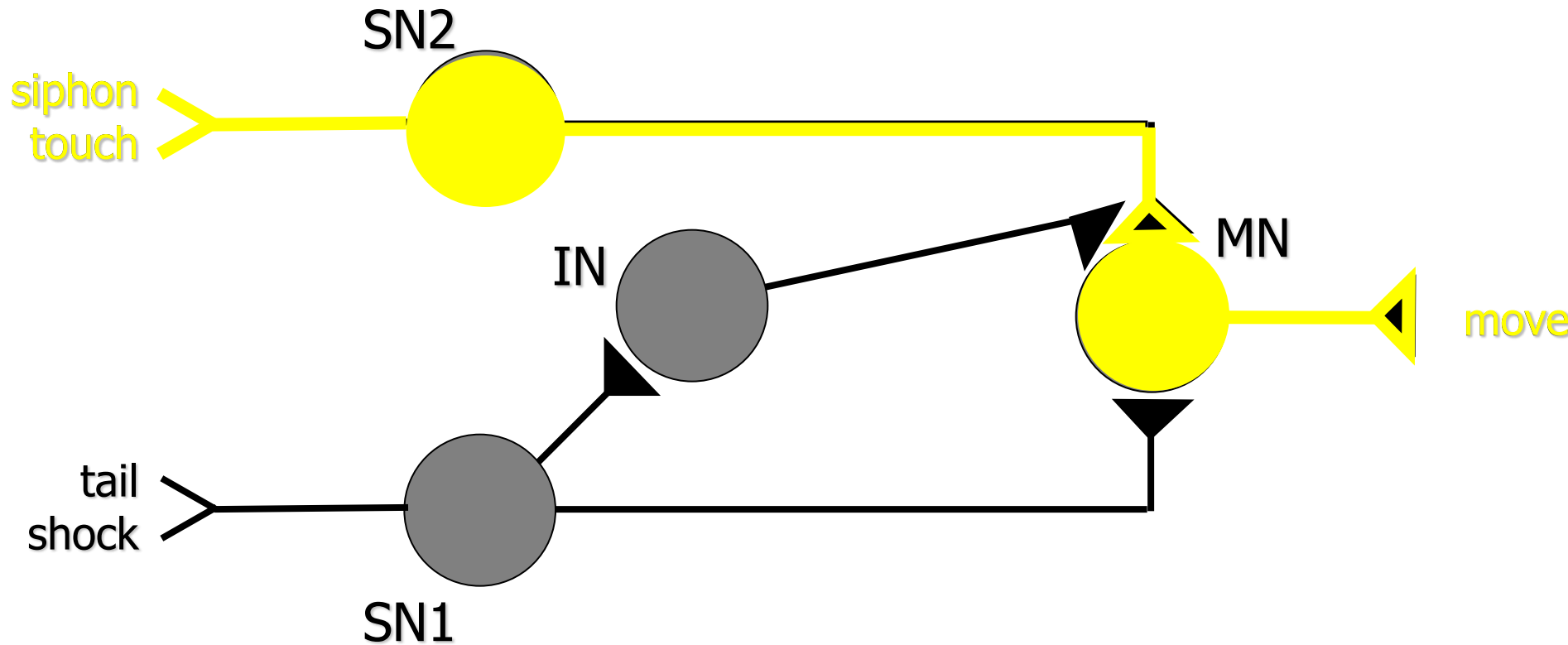
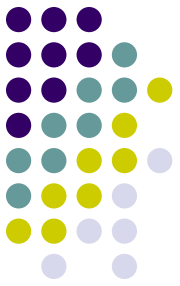
Neural Mechanism: Tail shock

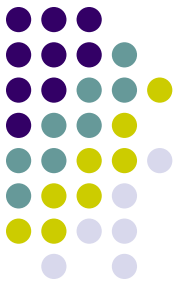


Neural Mechanism: learning



Neural Mechanism: Learned siphon touch response





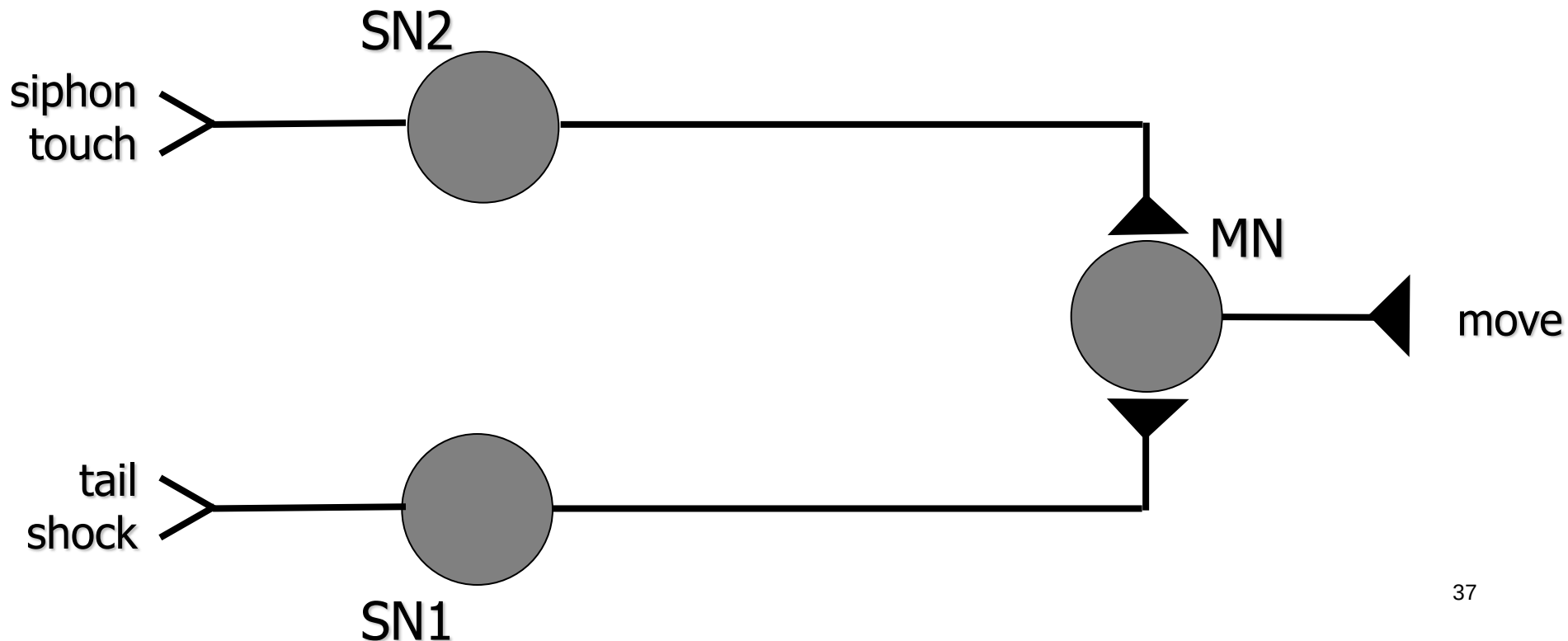
Simplified Learning Principle

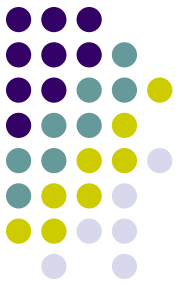
- If both SN2 and IN are activated simultaneously, this changes the synapse strength between SN2 and MN:
- It makes that in this synapse more neurotransmitter is produced if SN2 is activated.
- After a number of times this leads to the new situation that also activation of SN2 leads to activation of MN.



Modeling Aplysia learning

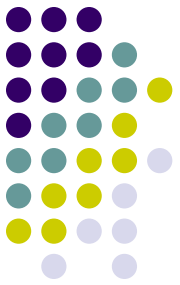
- Conceptualization
- Simplification: no intermediate neuron **IN**





Assignment

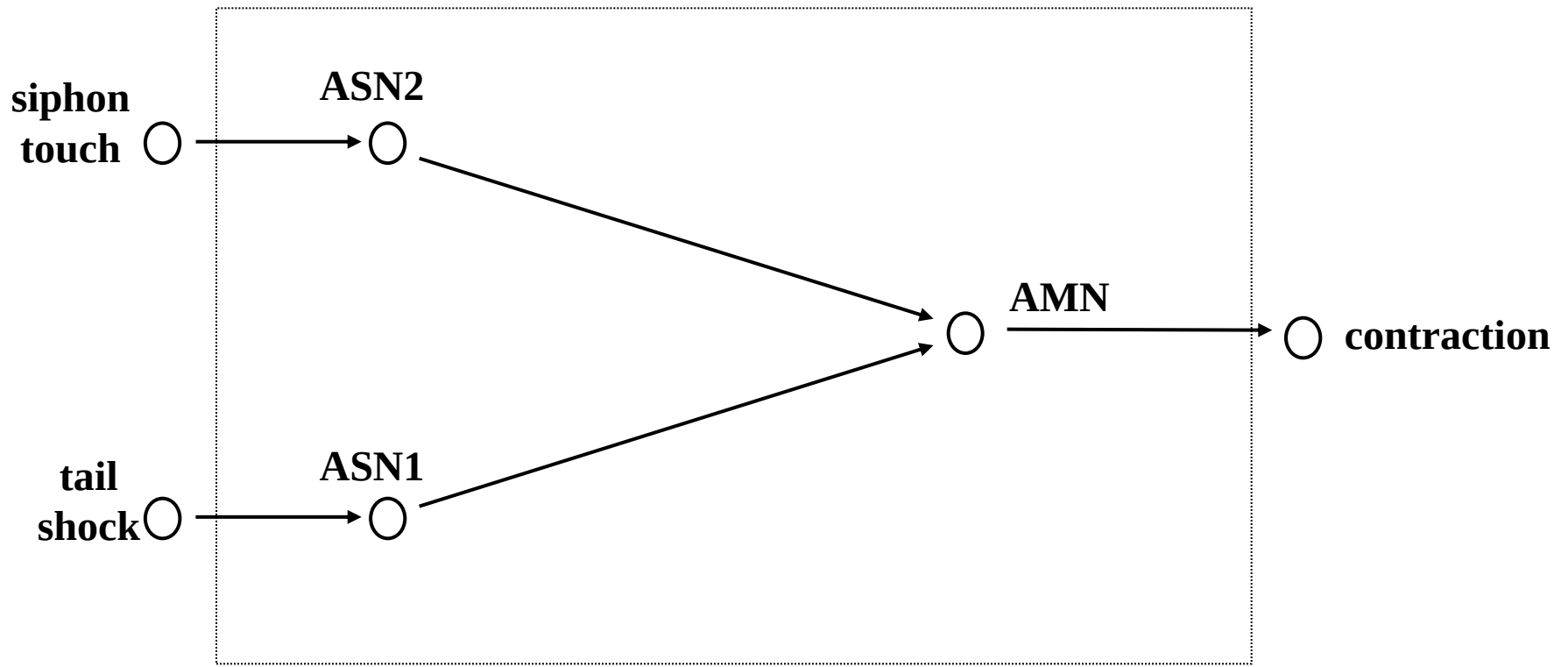
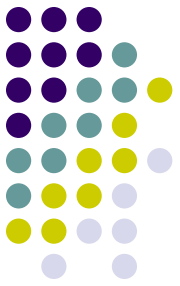
- Which concepts do you think are relevant?
 - Make a list of concepts
 - *3 minutes*
 - *groups of three; I will ask a few of you to come up with suggestions*
-
- Important question:
 - what do we consider as concepts?



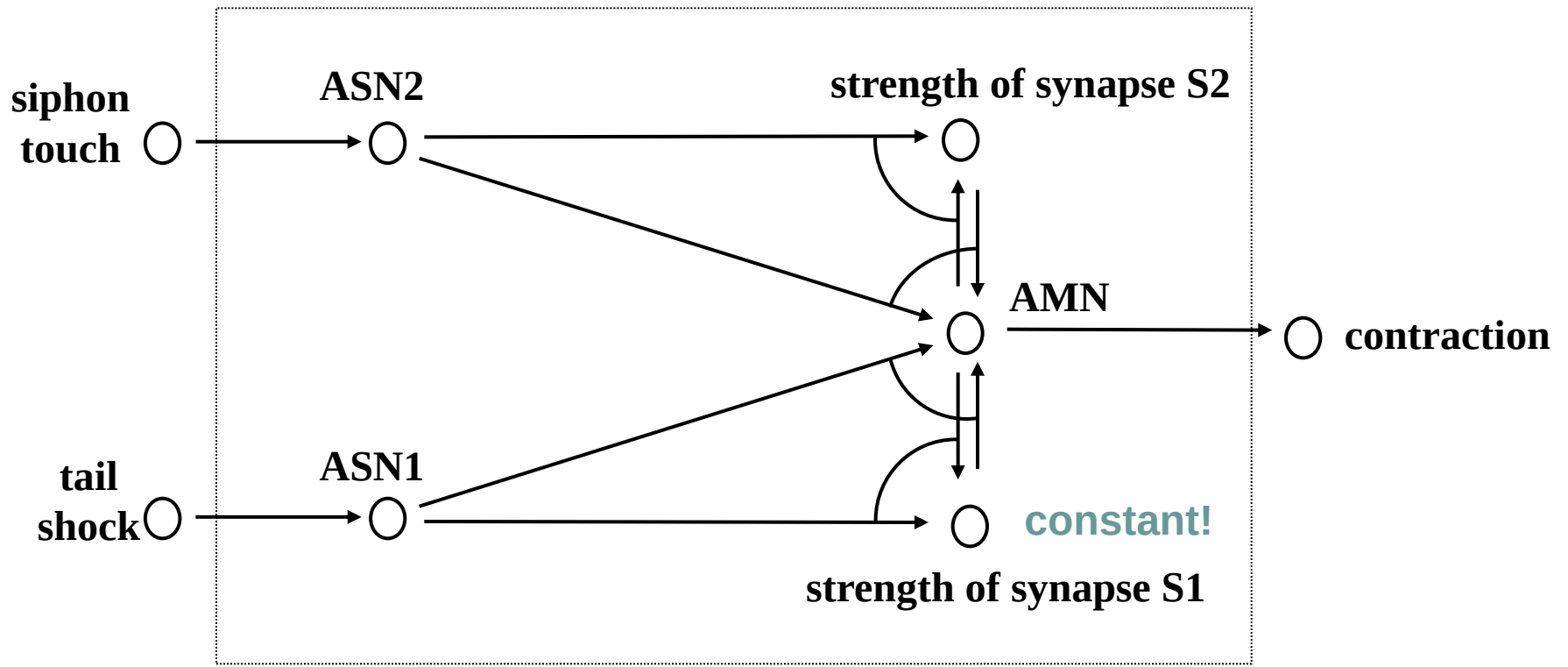
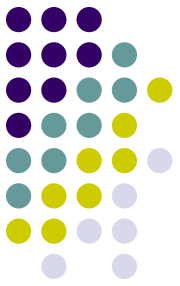
Concepts

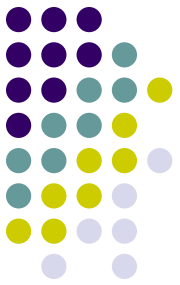
- tail shock
- siphon touch
- contraction
- sensory neuron SN1 is activated
- sensory neuron SN2 is activated
- motoneuron MN is activated

Relations between concepts



Relations between concepts





Relations

Sensory processes

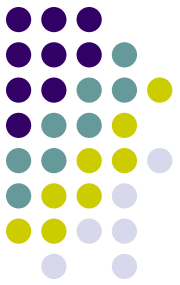
- A tail shock affects activation of SN1
- A siphon touch affects activation of SN2

Action generation

- Activation of SN1 affects activation of MN
- Activation of SN2 together with a certain strength of synapse S2 affects activation of MN
- Activation of MN affects contraction

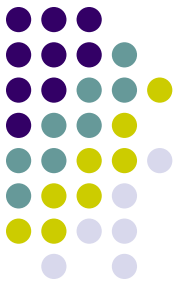
Adaptation

- Simultaneous activation of MN and SN2 affects the strength of synapse S2 between SN2 and MN by increasing it



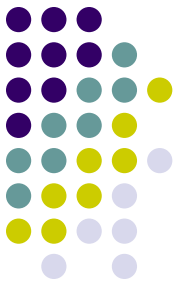
Expected patterns:

- Initially a contraction occurs after a shock, but no reaction upon a siphon touch
- After a training period of a number of trials (each of a siphon touch followed by a tail shock), contraction will take place after any siphon touch
- During a training period the strength of the synapse increases and eventually reaches a maximal strength



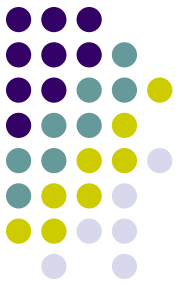
Formalization of concepts

- tail shock (*TailShock*)
- siphon touch (*SiphonTouch*)
- contraction (*Contraction*)
- sensory neuron SN1 is activated (*ASN1*)
- sensory neuron SN2 is activated (*ASN2*)
- motoneuron MN is activated (*AMN*)
- the synapse S1 between SN1 and MN has a certain strength (*S1*)
- the synapse S2 between SN2 and MN has a certain strength (*S2*)



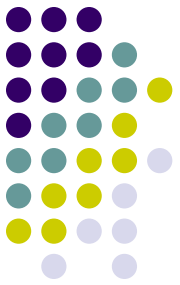
Formalization

- Concepts:
 - events and neuron activation: [0/1]
 - synapse strength: integer
- Relations
 - From tail shock to activation of SN1
 - $ASN1(t+\Delta t) = \max(TailShock(t), TailShock(t-\Delta t), TailShock(t-2\Delta t))$
 - From siphon touch to activation of SN2
 - $ASN2(t+\Delta t) = \max(SiphonTouch(t), SiphonTouch(t-\Delta t), SiphonTouch(t-2\Delta t))$



Formalization (2)

- From activation of MN to contraction
 - $Contraction(t+\Delta t) = AMN(t)$
- From activation of SN1 or SN2 to activation of MN
 - $AMN(t+\Delta t) = \begin{matrix} 1 & \text{if } ASN1(t) = 1 \text{ or:} \\ & S2(t) = 4 \text{ and } ASN2(t) = 1 \\ 0 & \text{else} \end{matrix}$
- Adjustment of the synapse strength based on simultaneous activation of MN and SN2
 - $S2(t+\Delta t) = \begin{matrix} S2(t) + 1 & \text{if } S2(t) < 4 \text{ and} \\ & AMN(t) = ASN2(t) = 1 \\ S2(t) & \text{else} \end{matrix}$



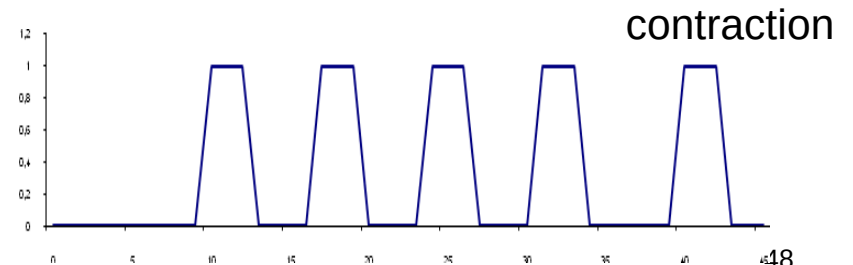
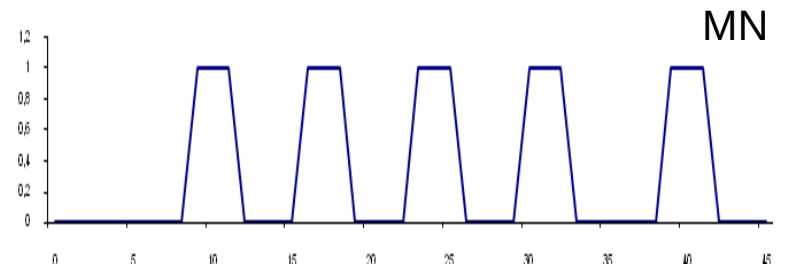
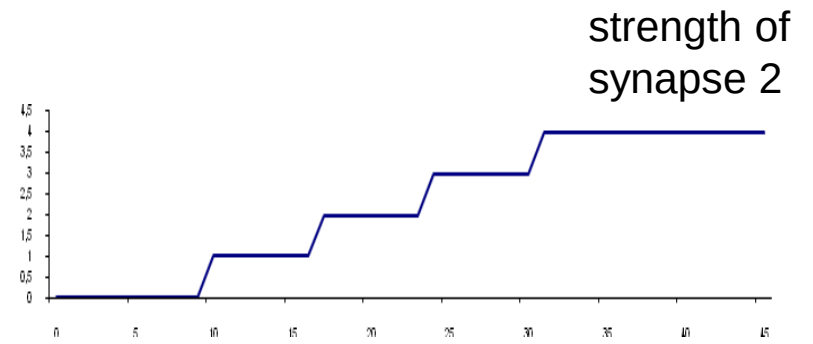
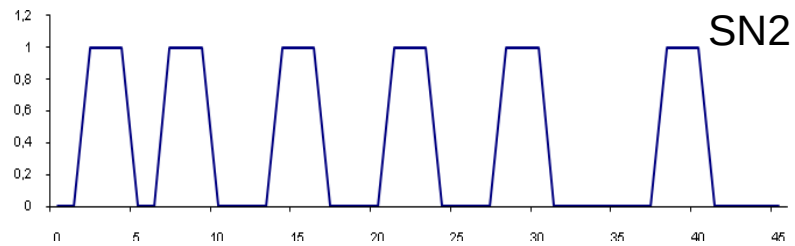
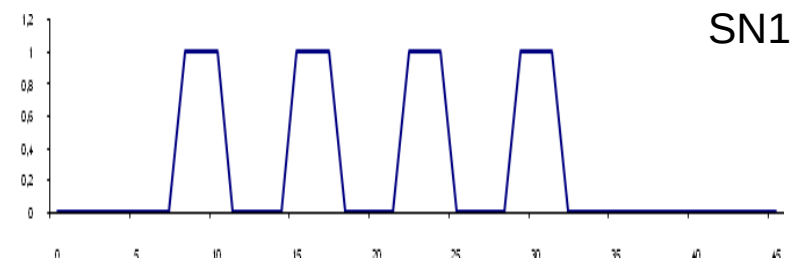
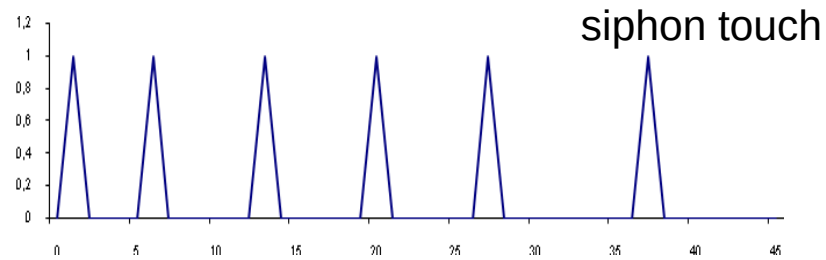
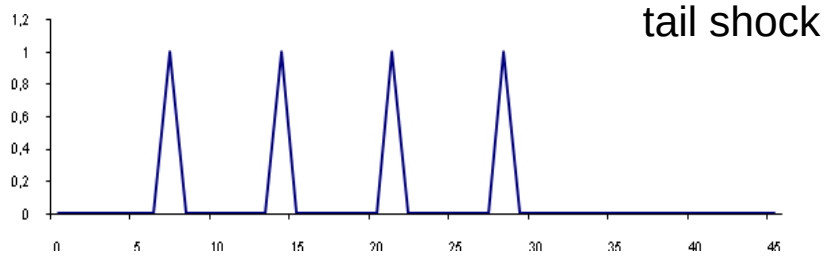
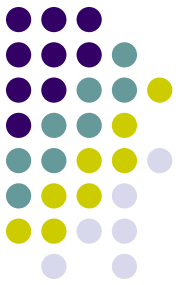
Temporal relations in Excel

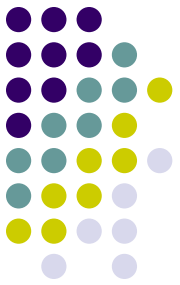
t	tail shock	siphon touch	ASN1	ASN2	AMN	synapse S2	contraction
0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
2	0	1	1	0	0	0	0
3	0	0	1	1	0	0	0
4	0	0	1	1	1	0	0
5	0	0	0	1	1	1	1

Diagram illustrating temporal relations in Excel. The table shows the state of various variables (tail shock, siphon touch, ASN1, ASN2, AMN, synapse S2, contraction) over time steps t from 0 to 5. Arrows indicate causal relationships between events:

- tail shock at $t=1$ leads to ASN1 at $t=2$.
- siphon touch at $t=2$ leads to ASN2 at $t=3$.
- ASN1 at $t=3$ leads to AMN at $t=4$.
- ASN2 at $t=4$ leads to synapse S2 at $t=5$.
- AMN at $t=4$ leads to synapse S2 at $t=5$.
- synapse S2 at $t=4$ leads to contraction at $t=5$.
- synapse S2 at $t=5$ leads to contraction at $t=5$.

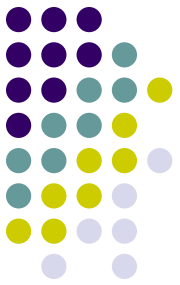
Simulations





Discussion

- Assumptions and choices in the model
 - The adjustments of the synapse strength are linear: every trial the same increase, until a maximum is reached.
 - There is no decay of synapse strength.
 - Activations of neurons last three time steps.
 - In a training period the stimuli (siphon touch and tail shock) are presented at two successive time steps.



Discussion (2)

- Variations:
 - Faster or slower learning.
 - Activation of the motoneuron MN by sensory neuron SN2 can take place in case the strength of the synapse connecting them is above a certain threshold, not at a maximum value.
 - The adjustments of the synapse strength can be taken not linear but bounded by a maximum.
 - Decay of synapse strength can be incorporated.
 - The activation periods of neurons may be taken to last longer or shorter than three time steps.
 - In a training period the time difference between the two stimuli (tail shock and siphon touch) can be varied.